MATHS MANTHAN
• TUITION CLASSES •
e-mail: contact@mathsmanthan.co.in | Website:- mathsmanthan.co.in

Subject: Mathematics | **CBSE CLASS 10 – PERIODIC ASSESSMENT TEST – 07** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 4: Quadratic Equations** | **MM: 30**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, zero skipped steps, and final boxed answers with proper units.
2. In alternative (**OR**) sub-questions, complete step-by-step solutions and equal step marking apply to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Roots rejected by physical domain/restrictions are explicitly stated.

Official Evaluation Key | Maths Manthan Meerut

Section A: MCQ & Assertion-Reason Questions

MATHS MANTHAN | $6 \times 1 = 6$ Marks

Q1. The least positive value of k for which the quadratic equation $2x^2 + kx - 4 = 0$ has rational roots is: [1]

(A) $\pm 2\sqrt{2}$ **(B)** 2
(C) 0 **(D)** $\sqrt{2}$

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 2

A quadratic equation $ax^2 + bx + c = 0$ with rational coefficients has rational roots if and only if its discriminant D is the square of a rational number:

Formula Used: $D = b^2 - 4ac = k^2 - 4(2)(-4) = k^2 + 32$

Testing the given options (note k must be **positive**, so $k > 0$):

- $k = 0$ is not positive, and gives $D = 32$ (roots $\pm\sqrt{2}$, irrational).
- $k = \sqrt{2} \implies D = 2 + 32 = 34$, not a square of a rational (and b itself is irrational).
- $k = 2 \implies D = 2^2 + 32 = 36 = 6^2$ (perfect square!), giving rational roots $x = \dfrac{-2 \pm 6}{4} = 1,\ -2$.

Notice $k = 1$ gives $D = 33$ (not a square), so $k = 2$ is indeed the least positive value. Hence, option **(B)** is correct. **[1 mark]**

Q2. If the sum of the roots of the equation $x^2 - (k+6)x + 2(2k-1) = 0$ is equal to half of their product, then $k =$: [1]

(A) 6 **(B)** 7
(C) 5 **(D)** -7

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 7

For $x^2 - (k+6)x + 2(2k-1) = 0$, we have $a = 1$, $b = -(k+6)$, and $c = 2(2k-1)$.

Formula Used: $\alpha + \beta = -\frac{b}{a}, \qquad \alpha\beta = \frac{c}{a}$

Here $\alpha + \beta = k + 6$ and $\alpha\beta = 2(2k-1)$. Given $\alpha + \beta = \frac{1}{2}\alpha\beta$:

$$k + 6 = \frac{1}{2}[2(2k-1)] \implies k + 6 = 2k - 1 \implies \boxed{k = 7}$$

Hence, option **(B)** is correct. **[1 mark]**

Q3. The discriminant of the quadratic equation $(x+5)^2 = 2(5x-3)$ is: [1]

(A) 124 **(B)** -124
(C) 31 **(D)** -31

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) -124

Rewriting $(x+5)^2 = 2(5x-3)$ in standard form $ax^2 + bx + c = 0$:

Formula Used: $D = b^2 - 4ac$

$$x^2 + 10x + 25 = 10x - 6 \implies x^2 + 0 \cdot x + 31 = 0$$

Here $a = 1$, $b = 0$, and $c = 31$. Thus:

$$D = (0)^2 - 4(1)(31) = \boxed{-124}$$

Hence, option **(B)** is correct. **[1 mark]**

Q4. If $y = 1$ is a common root of the equations $ay^2 + ay + 3 = 0$ and $y^2 + y + b = 0$, then ab equals: [1]

(A) 3 **(B)** -3
(C) 4 **(D)** 3.5

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) 3

Since $y = 1$ satisfies both equations:

Formula Used: Substitute $y = 1$ into both quadratic equations

- From $ay^2 + ay + 3 = 0$: $a(1)^2 + a(1) + 3 = 0 \implies 2a = -3 \implies a = -\frac{3}{2}$.
- From $y^2 + y + b = 0$: $(1)^2 + 1 + b = 0 \implies 2 + b = 0 \implies b = -2$.

Therefore, $ab = \left(-\frac{3}{2}\right)(-2) = \boxed{3}$. Hence, option **(A)** is correct. **[1 mark]**

Q5. If the quadratic equation $x^2 + 4x + k = 0$ has real and distinct roots, then: [1]

(A) $k < 4$ **(B)** $k > 4$

(C) $k \leq 4$ **(D)** $k \geq 4$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) $k < 4$

Comparing $x^2 + 4x + k = 0$ with $ax^2 + bx + c = 0$, we have $a = 1$, $b = 4$, and $c = k$.

Formula Used: Real and distinct roots $\iff D = b^2 - 4ac > 0$

Setting $D > 0$:

$$(4)^2 - 4(1)(k) > 0 \implies 16 - 4k > 0 \implies 16 > 4k \implies \boxed{k < 4}$$

Hence, option **(A)** is correct. **[1 mark]**

Q6. Assertion-Reason: [1]

Directions: *(A) Both A & R true, R explains A; (B) Both true, R does NOT explain A; (C) A true, R false; (D) A false, R true.*

Assertion (A): The roots of the quadratic equation $x^2 - 8x + 16 = 0$ are real and equal.

Reason (R): A quadratic equation $ax^2 + bx + c = 0$ has equal roots if its discriminant $D = b^2 - 4ac = 0$.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) Both Assertion (A) and Reason (R) are true, and Reason (R) is the correct explanation of Assertion (A).

For $x^2 - 8x + 16 = 0$, we have $a = 1$, $b = -8$, and $c = 16$.

Formula Used: $D = b^2 - 4ac = 0 \implies$ real and equal roots

$$D = (-8)^2 - 4(1)(16) = 64 - 64 = 0$$

Since $D = 0$, the equation $x^2 - 8x + 16 = (x - 4)^2 = 0$ has equal real roots 4, 4. Thus Assertion (A) is **true**, Reason (R) is **true**, and Reason (R) correctly explains Assertion (A). **[1 mark]**

Section B: Very Short Answer Questions

MATHS MANTHAN | $2 \times 2 = 4$ Marks

Q7. Find the values of k for which the quadratic equation $kx^2 - 6x - 2 = 0$ has real roots. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

For $kx^2 - 6x - 2 = 0$ to be a quadratic equation, $k \neq 0$. Here $a = k$, $b = -6$, and $c = -2$.

Formula Used: Real roots $\iff D = b^2 - 4ac \geq 0$

Computing D and setting $D \geq 0$:

$$(-6)^2 - 4(k)(-2) \geq 0 \implies 36 + 8k \geq 0 \implies 8k \geq -36 \implies k \geq -\frac{36}{8} \implies k \geq -\frac{9}{2}$$ **[1½ marks]**

Since the equation must remain quadratic, $k \neq 0$ (at $k = 0$ it becomes $-6x - 2 = 0$, linear).

Hence, the equation has real roots for $\boxed{k \geq -\frac{9}{2},\ k \neq 0}$. **[½ mark]**

Q8. Solve the following quadratic equation for x by factorization: $3x^2 - 2\sqrt{6}x + 2 = 0$. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given equation: $3x^2 - 2\sqrt{6}x + 2 = 0$. Here $a = 3$, $b = -2\sqrt{6}$, and $c = 2$.

Formula Used: $ac = 3 \times 2 = 6 = (-\sqrt{6})(-\sqrt{6}), \qquad (-\sqrt{6}) + (-\sqrt{6}) = -2\sqrt{6}$

Splitting the middle term as $-2\sqrt{6}x = -\sqrt{6}x - \sqrt{6}x$:

$$3x^2 - \sqrt{6}x - \sqrt{6}x + 2 = 0$$
$$\sqrt{3}x(\sqrt{3}x - \sqrt{2}) - \sqrt{2}(\sqrt{3}x - \sqrt{2}) = 0 \quad \textbf{[1 mark]}$$
$$(\sqrt{3}x - \sqrt{2})^2 = 0$$

Equating the factor to zero gives two equal real roots:

$$\sqrt{3}x - \sqrt{2} = 0 \implies x = \frac{\sqrt{2}}{\sqrt{3}} = \frac{\sqrt{6}}{3}$$

Hence, the roots are $\boxed{x = \frac{\sqrt{6}}{3}, \frac{\sqrt{6}}{3}}$ $\left(\text{or } \sqrt{\frac{2}{3}}\right)$. **[1 mark]**

Section C: Short Answer Questions

MATHS MANTHAN | $2 \times 3 = 6$ Marks

Q9. Solve the following equation for x: $\dfrac{x-1}{x-2} + \dfrac{x-3}{x-4} = 3\dfrac{1}{3}, \quad x \neq 2,\ 4.$ **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Converting the mixed fraction $3\frac{1}{3} = \frac{10}{3}$ and taking the LCM on the LHS:

Formula Used: $\dfrac{x-1}{x-2} + \dfrac{x-3}{x-4} = \dfrac{(x-1)(x-4) + (x-3)(x-2)}{(x-2)(x-4)}$

$$\frac{(x^2 - 5x + 4) + (x^2 - 5x + 6)}{x^2 - 6x + 8} = \frac{10}{3}$$
$$\frac{2x^2 - 10x + 10}{x^2 - 6x + 8} = \frac{10}{3} \quad \textbf{[1 mark]}$$

Dividing both sides by 2:

$$\frac{x^2 - 5x + 5}{x^2 - 6x + 8} = \frac{5}{3} \implies 3x^2 - 15x + 15 = 5x^2 - 30x + 40 \implies 2x^2 - 15x + 25 = 0 \quad \textbf{[1 mark]}$$

Factorising ($2 \times 25 = 50$; $-10 - 5 = -15$):

$$2x^2 - 10x - 5x + 25 = 0 \implies (x-5)(2x-5) = 0 \implies x = 5 \text{ or } x = \frac{5}{2}$$

Both values satisfy $x \neq 2, 4$. Hence, $\boxed{x = 5 \text{ and } x = \frac{5}{2}}$. **[1 mark]**

Q10. Solve the following equation for x: $\dfrac{x+3}{x+2} = \dfrac{3x-7}{2x-3}, \quad x \neq -2, \dfrac{3}{2}$. [3]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Given equation: $\dfrac{x+3}{x+2} = \dfrac{3x-7}{2x-3}$, where $x \neq -2, \dfrac{3}{2}$.

Formula Used: $\dfrac{A}{B} = \dfrac{C}{D} \implies A \cdot D = B \cdot C \quad (B, D \neq 0)$

Cross-multiplying:

$$(x+3)(2x-3) = (3x-7)(x+2)$$
$$2x^2 - 3x + 6x - 9 = 3x^2 + 6x - 7x - 14 \quad \textbf{[1 mark]}$$
$$2x^2 + 3x - 9 = 3x^2 - x - 14$$
$$x^2 - 4x - 5 = 0 \quad \textbf{[1 mark]}$$

Factorising ($-5 \times 1 = -5$ and $-5 + 1 = -4$):

$$x^2 - 5x + x - 5 = 0 \implies x(x-5) + 1(x-5) = 0 \implies (x-5)(x+1) = 0$$

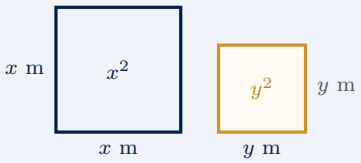
This gives $x = 5$ or $x = -1$. Both roots satisfy the domain constraints ($x \neq -2, \frac{3}{2}$).
Hence, the solutions are $\boxed{x = 5 \text{ and } x = -1}$. **[1 mark]**

Section D: Long Answer / HOTS Questions

MATHS MANTHAN | $2 \times 5 = 10$ Marks

Q11. The sum of the areas of two squares is 468 m^2. If the difference of their perimeters is 24 m, formulate the quadratic equation and determine the lengths of the sides of the two squares. [5]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]



Let the side of the larger square be x m and that of the smaller square be y m ($x > y > 0$), as depicted alongside. Their respective areas are x^2 and y^2, and their perimeters are $4x$ and $4y$.

Formula Used: Area of square = (side)^2, Perimeter of square = $4 \times$ side

Given the difference of their perimeters is 24 m:

$$4x - 4y = 24 \implies x - y = 6 \implies x = y + 6 \quad \textbf{[1 mark]}$$

Also, the sum of their areas is 468 m^2:

$$x^2 + y^2 = 468$$

Substituting $x = y + 6$:

$$(y+6)^2 + y^2 = 468$$
$$(y^2 + 12y + 36) + y^2 = 468 \implies 2y^2 + 12y - 432 = 0 \quad \textbf{[1}\tfrac{1}{2}\textbf{ marks]}$$
$$y^2 + 6y - 216 = 0 \quad \text{(required quadratic equation, dividing by 2)} \quad \textbf{[}\tfrac{1}{2}\textbf{ mark]}$$

Factorising $y^2 + 6y - 216 = 0$ ($18 \times (-12) = -216$ and $18 - 12 = 6$):

$$y^2 + 18y - 12y - 216 = 0 \implies y(y+18) - 12(y+18) = 0 \implies (y+18)(y-12) = 0 \quad \textbf{[1 mark]}$$

Thus $y = -18$ or $y = 12$. Since side length cannot be negative ($y > 0$), we reject $y = -18$.
Therefore, $y = 12$ m and $x = 12 + 6 = 18$ m. (Verification: $18^2 + 12^2 = 324 + 144 = 468$.)
Hence, the sides of the two squares are $\boxed{18 \text{ m and } 12 \text{ m}}$. **[1 mark]**

Q12. A train travels a distance of 480 km at a uniform speed. If the speed had been 8 km/h less, it would have taken 3 hours more to cover the same distance. Find the original uniform speed of the train. [5]

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

Let the original uniform speed of the train be x km/h ($x > 8$).
Total distance = 480 km. Reduced speed = $(x - 8)$ km/h.

Formula Used: $\text{Time} = \dfrac{\text{Distance}}{\text{Speed}}$, $\quad t_{\text{slow}} - t_{\text{original}} = 3$ hours

Time at original speed = $\dfrac{480}{x}$ hours; Time at reduced speed = $\dfrac{480}{x-8}$ hours. **[1 mark]**

By the problem condition:

$$\frac{480}{x-8} - \frac{480}{x} = 3$$

$$480\left[\frac{x-(x-8)}{x(x-8)}\right] = 3 \implies \frac{480 \times 8}{x^2 - 8x} = 3 \quad \textbf{[1}\tfrac{1}{2}\textbf{ marks]}$$

Dividing both sides by 3:

$$\frac{160 \times 8}{x^2 - 8x} = 1 \implies x^2 - 8x = 1280 \implies x^2 - 8x - 1280 = 0 \quad \textbf{[1 mark]}$$

Factorising by splitting the middle term ($-40 \times 32 = -1280$ and $-40 + 32 = -8$):

$$x^2 - 40x + 32x - 1280 = 0 \implies x(x-40) + 32(x-40) = 0$$
$$\implies (x-40)(x+32) = 0 \implies x = 40 \text{ or } x = -32 \quad \textbf{[1 mark]}$$

Since speed cannot be negative ($x > 0$), we reject $x = -32$. (Check: $\frac{480}{32} - \frac{480}{40} = 15 - 12 = 3$ h.)
Hence, the original uniform speed of the train is $\boxed{40 \text{ km/h}}$. **[$\frac{1}{2}$ mark]**

Q13. Case Study: Architectural Interior Design & Floor Tiling Optimization

A banquet hall has a large rectangular floor that can be completely covered with 200 identical square ceramic tiles of side length x units. During an interior redesign, the project architect discovers that if the side length of each tile is increased by 1 unit, it would require only 128 tiles to pave the entire floor surface completely. [4]

(i) Express the total surface area of the floor in terms of x when tiled using 200 square tiles. **[1]**

(ii) Set up the quadratic equation in x in standard form equating the two area expressions. **[1]**

(iii) Find the original side length x of each tile, and calculate the total surface area of the floor. **OR** Determine how many tiles would be needed to pave the floor if tiles of side length $(x+2)$ units were used. **[2]**

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]



200 tiles (x) vs 128 tiles ($x+1$)

As shown in the comparison schematic alongside, the same rectangular floor of fixed area is completely covered either by 200 smaller square tiles of side x units or by 128 larger square tiles of side $(x+1)$ units.

We know the area covered equals the number of tiles times the area of each square tile:

Formula Used: Total Floor Area = (Number of tiles) × (Side of tile)^2

(i) Total Floor Area Expression: Each square tile has side x units and area x^2 sq. units.

$$\text{Total floor area with 200 tiles} = \boxed{200x^2 \text{ sq. units}} \quad \textbf{[1 mark]}$$

(ii) Quadratic Equation in Standard Form: With 128 tiles of side $(x+1)$ units, total area $= 128(x+1)^2$. Equating both expressions for the same floor area:

$$\begin{aligned} 200x^2 &= 128(x+1)^2 \\ 25x^2 &= 16(x^2+2x+1) \quad \text{(dividing both sides by 8)} \\ 25x^2 &= 16x^2+32x+16 \implies \boxed{9x^2-32x-16=0} \end{aligned}$$ **[1 mark]**

(iii) First Part (Tile Side x & Floor Area): Factorising $(9 \times (-16) = -144;\ -36+4 = -32)$:

$$9x^2-36x+4x-16=0 \implies 9x(x-4)+4(x-4)=0 \implies (x-4)(9x+4)=0 \implies x=4 \text{ or } x=-\frac{4}{9}$$

Since length cannot be negative $(x>0)$, we reject $x=-\frac{4}{9}$.

So $x=4$ units, and total floor area $= 200(4)^2 = 200 \times 16 = 3200$ sq. units.

Hence, $\boxed{\text{Original tile side } x=4 \text{ units}}$ and $\boxed{\text{Floor area} = 3200 \text{ sq. units}}$. **[2 marks]**

OR (Alternative Part — Tiles of Side $(x+2)$): With $x=4$, new tile side $= x+2 = 4+2 = 6$ units, so area per tile $= 6^2 = 36$ sq. units.

$$\text{Exact number of tiles} = \frac{\text{Total floor area}}{\text{Area of one tile}} = \frac{3200}{36} = \frac{800}{9} = 88\frac{8}{9} \approx 88.89 \text{ tiles}$$

Thus $\boxed{\frac{800}{9} = 88\frac{8}{9} \text{ exact tiles}}$ (i.e., $\boxed{89 \text{ whole tiles}}$ needed to cover completely). **[2 marks]**

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★